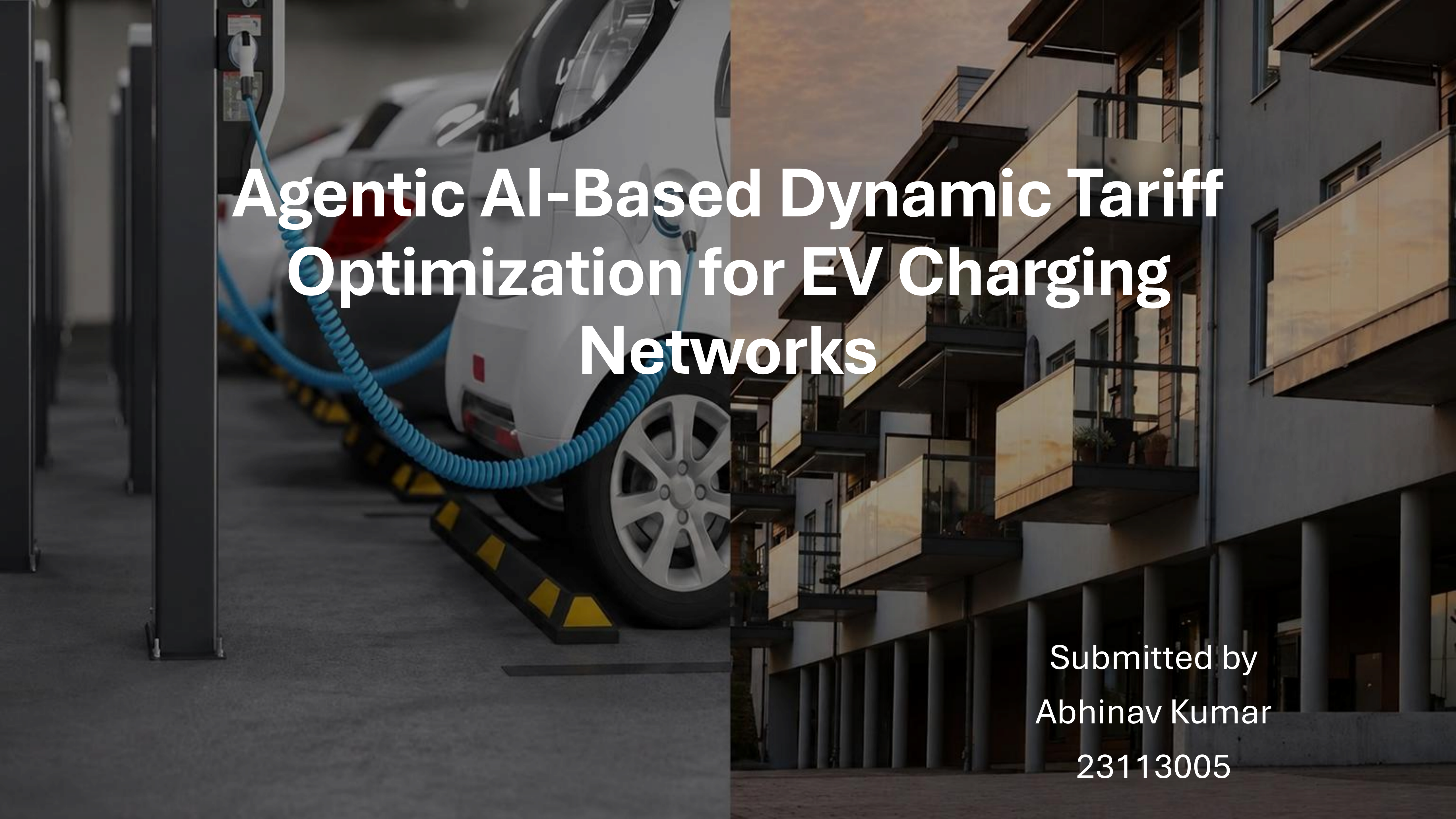

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Submitted by

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Data Landscape and Preprocessing

• **ACN-Data:** Session level charging records from Caltech/JPL. It is used for timestamps, energy delivered, duration, station and user behavior.

• **UrbanEV / ST-EVCDP:** 5-minute station level data from Shenzhen. It is used for occupancy, volume, duration, price, and spatial patterns.

•**Preprocessing decisions:**

• I have cleaned timestamps and removed incomplete session records.

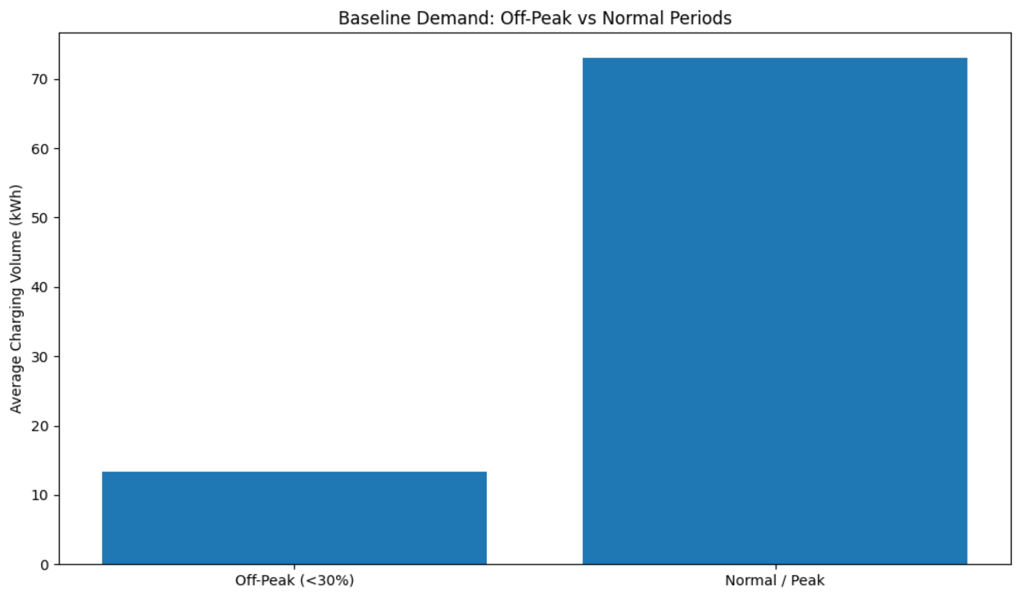
• I have aligned UrbanEV files by time_idx and grid.

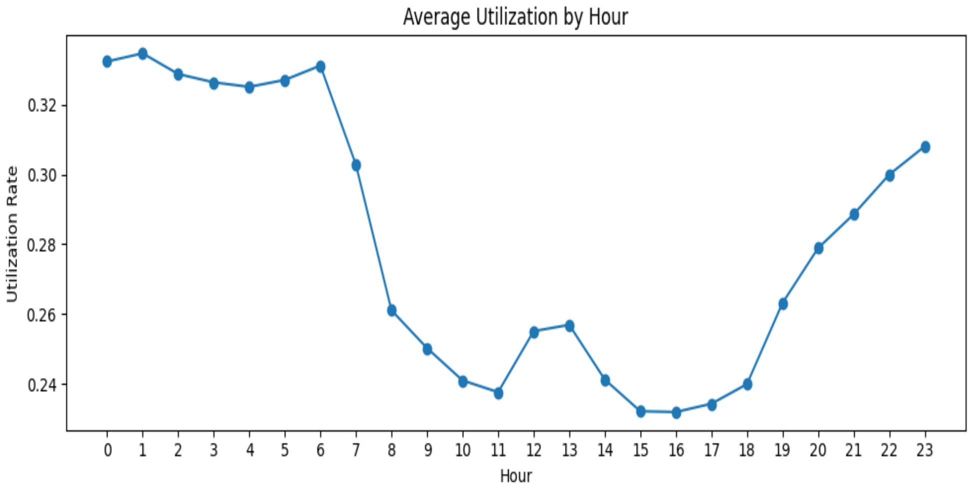
• I engineered utilization, revenue proxy, queue proxy, and occupancy density.

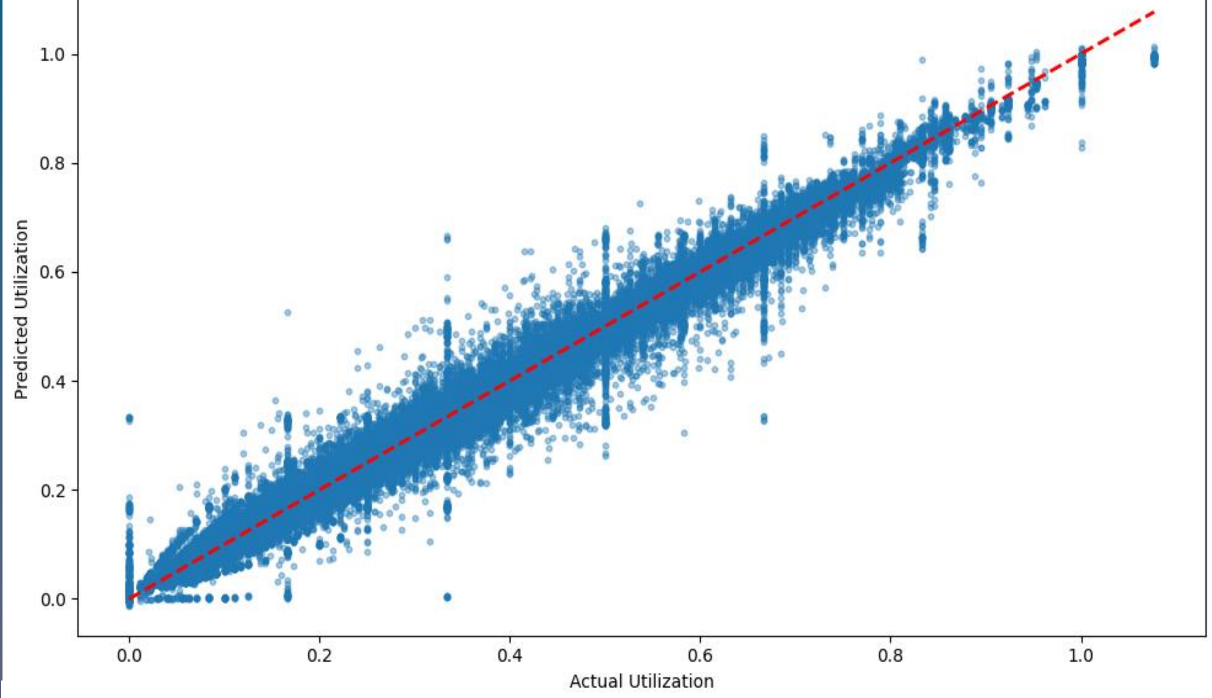
• I also added lag and rolling features for forecasting.

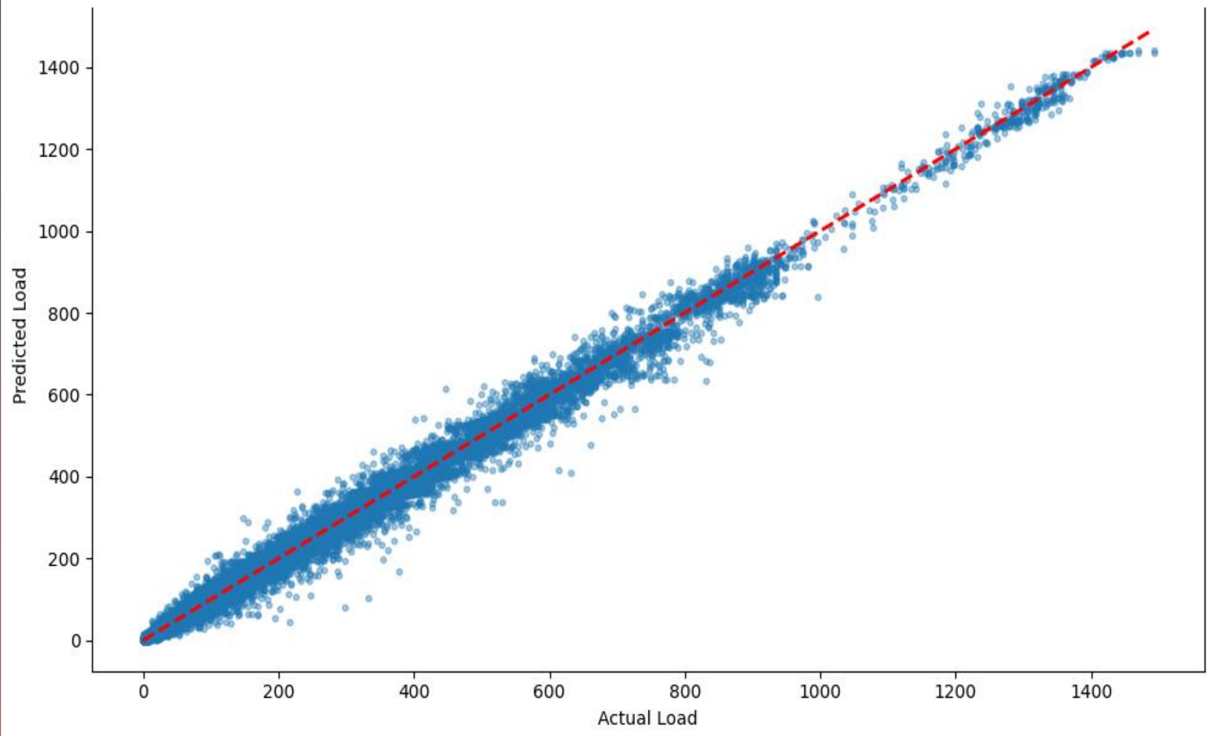
Temporal and Spatial Charging Patterns

- Charging demand varies strongly by hour of day and day of week
- Utilization is generally low on average, but peak windows create localized congestion
- Off-peak periods show clear room for discount-based demand shifting
- Spatial differences exist across stations, with some grids persistently more active than others
- Adjacent stations show related demand patterns, supporting spatial modeling









Demand Prediction Modeling and Result

- Model used: **XGBoost Regressor**

- Targets : next-step utilization rate
  next-step charging load

- Feature set: lag features
  rolling means and standard deviations
  station metadata
  spatial neighbor features
  time-of-day and weekday cyclical encodings

- Results: Utilization $R^2 = 0.992$
  Load $R^2 = 0.9962$

- Interpretation: The model captures short-term demand dynamics very well

Utilization Forecast

[29]
0s

```
print("R2  :", round(r2_util, 4))
print("MAE :", round(mae_util, 4))
print("RMSE:", round(rmse_util, 4))
```

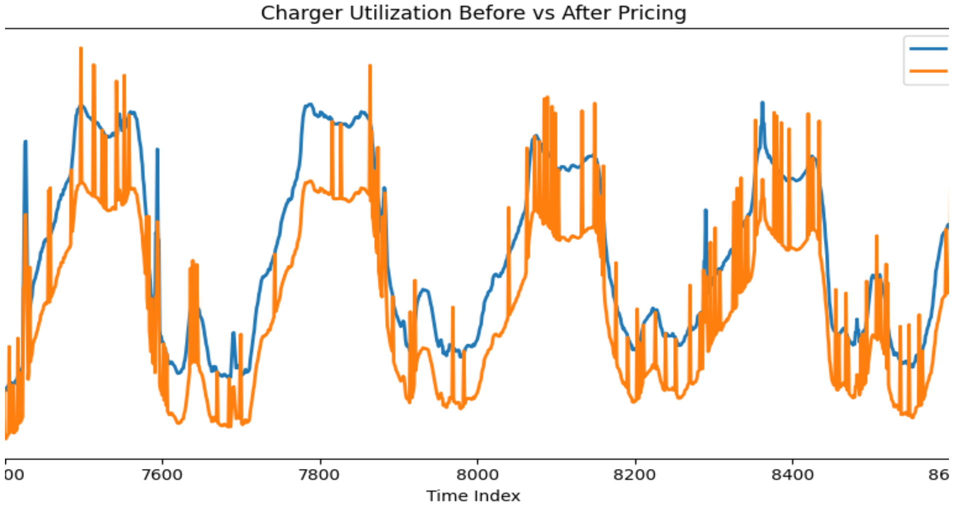
```
R2  : 0.992
MAE : 0.0071
RMSE: 0.0158
```

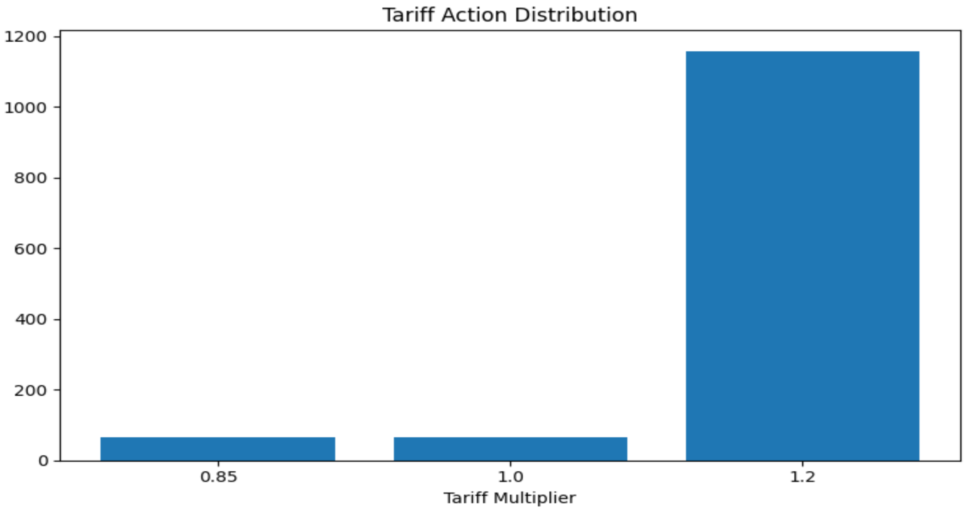
Charging Load Forecast

[31]
0s

```
print("R2  :", round(r2_load, 4))
print("MAE :", round(mae_load, 4))
print("RMSE:", round(rmse_load, 4))
```

```
R2  : 0.9962
MAE : 1.6922
RMSE: 42.6675
```

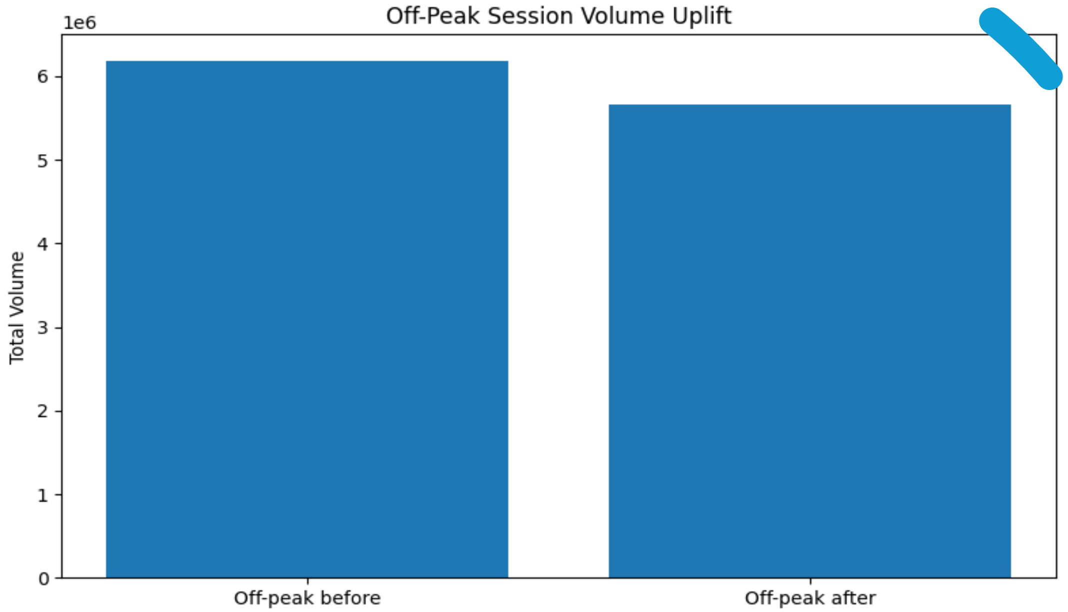


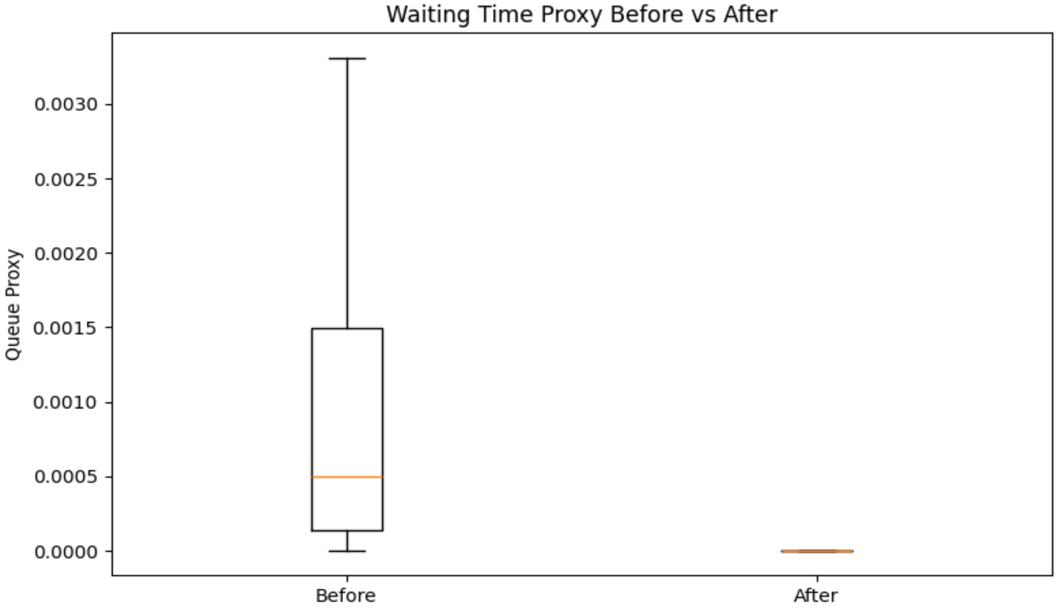


Dynamic Tariff Optimization Logic and Outcomes

- Baseline tariff: **₹15/kWh**
- Tariff logic: utilization > 80% → surge pricing
  utilization < 30% → discount pricing
  otherwise → base tariff
- Decision engine translates predicted demand into tariff actions.
- Observed outcomes: Revenue Gain**: 6.04%**
  Pricing Efficiency improved from **₹15.0 to ₹17.31/kWh**
  Off-Peak Uplift**: 29.48%**

- **Note:** Pricing outcomes are evaluated as simulated operational effects, not causal proof





Learning, Evaluation, and Operational Feedback

• Monitoring agent tracks: revenue gain
utilization before/after pricing
off-peak uplift
waiting-time proxy reduction
customer response rate
pricing efficiency score

• Results from the evaluation run:
waiting time proxy reduced to near zero.
feedback loop updates tariff action values over time.
customer response rate was negative under higher prices,
under higher prices, showing price sensitivity.

Business, Operational, and Policy Implications

What the Results Mean for EV Charging Networks?

• Dynamic pricing can improve **revenue efficiency** without relying on static flat tariffs
• Discounts help shift demand toward underused periods
• Forecasting can support station planning, congestion reduction, and load balancing
• Low average utilization suggests a need for pricing strategies that stimulate demand rather than only suppress peaks
• Policy and operational caution:
• Avoid overclaiming causality
• Results depend on assumptions about elasticity and simulated responses
• Fairness, user acceptance, and transparency matter in deployment


Thank You